

A Four-Adic Endpoint Packing Proposal for the Riemann Hypothesis

Positive parabolic atoms, logarithmic blocker chains, and the native von Mangoldt dual

ChatGPT Pro (GPT-5.6 Pro)

Primary mathematical synthesis and manuscript

Gideon Freund

Research harness, prompting, and project direction

Agentic Polymath Project

August 16, 2026

Scientific status. This document is an *unconditional proof proposal*, not an accepted proof of the Riemann Hypothesis. “Unconditional” means that the displayed implication does not assume RH, a zero-density estimate, a Mertens square-root bound, an unproved prime-correlation estimate, or any separately named RH-equivalent producer. The only new load-bearing step is the logarithmic blocker-chain estimate in theorem 5.3; a proof is supplied rather than assumed, but it has not been independently reconstructed. An error in that proof would invalidate the proposal. RH must therefore continue to be regarded as unproved unless and until hostile independent review accepts every interface.

Abstract

We give a self-contained proof proposal for the Riemann Hypothesis built from a research reset across the `gfreund123/riemann` repository. The proposal discards the factor-67 source compiler, the uncertified Target–Lorenz tail, the Q4 balanced Type-II hypothesis, the Cauchy–Jordan first-chaos domination hypothesis, and the finite Brownian stability finishes. It retains three exact ingredients: the positive parabolic endpoint atoms, their strictly positive radix-four detail responses, and the sparse positive von Mangoldt dual

$$Y_4(q) = \sum_{k=0}^{v_4(q)} 2^k \Lambda(q/4^k).$$

For every endpoint X , a backward triangular greedy constructs a nonnegative physical row d_X with all ordinary and radix-four capacities feasible. The repository previously reduced the remaining problem to the support of the final slack. The new step is a blocker-chain estimate

$$\hat{\lambda}_T - \lambda_T \leq 96 \log(2X),$$

proved from an exact first-crossing comparison for the endpoint-detail kernel. Since the diagonal response is at most $1/(2q^{3/2})$ and

$$\sum_{q \geq 2} \frac{Y_4(q)}{q^{3/2}} < 11,$$

the exact native deficit obeys

$$0 \leq J_\Lambda(X) - \mathcal{H}(d_X) < 528 \log(2X) = o(\log^2 X).$$

We then reproduce the one-sided endpoint dual, the prime-square moat, the Mellin transform, and the Landau one-sign argument, obtaining the candidate implication to RH. The paper includes a dependency ledger, route-by-route rejection map, explicit falsifiers, and a deterministic finite replay. The most delicate identity is isolated as equation (34); reviewers should attack it first.

Contents

1	Research reset and selection of the proof architecture	3
1.1	Routes inspected and their controlling obstacle	3
1.2	Provenance freeze	4
2	The native endpoint problem	4
2.1	Ordinary and radix-four capacities	4
2.2	The native von Mangoldt dual	5
2.3	Exact support and summability of the dual	5
3	Positive parabolic endpoint atoms	6
3.1	The scale frame	6
4	The exact finite compiler and its blocker identity	7
5	The new logarithmic blocker-chain theorem	7
5.1	Normalized first-crossing ratios	8
6	Uniform native-deficit estimate	9
7	From native deficit to the Riemann Hypothesis	10
7.1	The endpoint defect	10
7.2	Prime-square moat	10
7.3	Mellin transform and Landau converse	11
8	Adversarial interface audit	11
8.1	Type and normalization firewalls	11
8.2	Immediate mathematical falsifiers	12
8.3	Known non-proofs deliberately excluded	12
9	Finite replay and reconnaissance	12
10	Theorem and dependency ledger	13
11	Conclusion	14
A	Derivation of the sparse dual support	14
B	A reviewer-oriented expansion of the first-crossing calculation	14
C	Deterministic build and replay	15
D	Repository files consulted	15

1 Research reset and selection of the proof architecture

The repository contains a large number of exact finite identities, conditional criteria, failed full compositions, computational certificates, and later refutations. A proof reset must distinguish four different things:

- (i) a correct finite identity;
- (ii) an unconditional asymptotic estimate;
- (iii) an RH-equivalent producer stated as a theorem target;
- (iv) a complete proof composition whose interfaces use the same normalization and the same physical object.

A short route is not necessarily close to a proof if its final short statement is merely RH in new coordinates.

1.1 Routes inspected and their controlling obstacle

Family	Strongest retained mechanism	Reason not used as the closing producer
Factor-67 SONTR	/ Exact subcritical coefficient mass, native-detail duality, finite Hall and causal algebra	Repeated source-normalization failures: signed observations promoted to positive source, negative oriented child coordinate, activation knots, and uncertified transcendental intervals.
Target–Lorenz	Strong compact directed margins and a useful ordered-vector primal/dual	The published tail used singleton floating-point intervals for irrational square roots and logarithms; the live source-to-leaf common parent was not simultaneously certified.
Centered Q4 cubic	A zero-safe mean-zero cubic scalar, exact Mellin multiplier, prime-only reduction, and positive Peano potential	The balanced large-divisor Möbius/Type-II estimate remains RH-bearing. Generic Cauchy or large-sieve bounds recombine $\mu * \log = \Lambda$.
First-Hermite heat	Exact heat-wavelet bridges and phase-locked finite-difference gains	The fixed positive improvement beyond the constant-four frontier requires signed arithmetic not supplied by the scalar absolute majorant.
Cauchy–Jordan Clark	/ Positive Jordan sources, compound-Poisson semigroup, exact cocycles, and scattering factorizations	Boundary amplitude unitarity does not imply source curvature domination. The remaining first-chaos inequality is itself RH-equivalent.
Brownian mite	/ Her- mite Exact finite gamma/Dirichlet algebra and compact-height approximation	Raw all-large finite stability is refuted by high-frequency Bohr zeros; current symmetrized mixtures inherit the instability.

Family	Strongest retained mechanism	Reason not used as the closing producer
Pick / Loewner / Stieltjes	Exact finite necessary matrices and positive finite boxes	Finite positivity excludes only a finite feature box. No complete positivity class or authenticated negative witness is present.
Robin / Nicolas	Exact finite barriers and canonical reductions	The unbounded canonical tail remains uncontrolled.
Endpoint / occupancy	One-sided endpoint consumer, prime-square moat, and positive packing geometry	This family has a direct positive finite compiler; its only remaining object is a scalar slack of an explicit triangular system.

The endpoint-detail architecture is selected because it has the smallest number of type conversions. Its source atoms are already nonnegative physical rows. One coefficient is used in the row, ordinary columns, radix-four columns, score, and endpoint deficit. There are no rough children, no activation cells, no finite/continuum source identification, and no matrix port.

1.2 Provenance freeze

The principal source heads are:

Object	Source	Frozen SHA
Native dual and endpoint consumer	PR #468	9e2ae6d26a7920055e1f11327dc0ddeb7b855c61
Parabolic scale frame and defect transport	PR #265	e0ad1484d70098eb7c8be17655ea08568d5e2cd2
Endpoint-detail compiler	PR #530	6c818a35094b978863a08bde4227d1f4ad9b65d4
Sparse dual arithmetic	PR #477	5acd9007b4f4bb1792466f5013c39bf4eac33f9e

All claims imported from those heads are restated below. The new blocker theorem is not inherited from any branch.

2 The native endpoint problem

2.1 Ordinary and radix-four capacities

For an integer endpoint $X \geq 3$ and $q \geq 2$, define

$$w_X(q) = \frac{1}{\sqrt{q}} \log \frac{X}{q} \mathbf{1}_{q \leq X}, \quad \Omega_X(q) = w_X(q) - 2w_X(4q). \quad (1)$$

Then $\Omega_X(q) \geq 0$. More explicitly,

$$\Omega_X(q) = \frac{\log 4}{\sqrt{q}} \quad (2 \leq q \leq X/4), \quad \Omega_X(q) = w_X(q) \quad (X/4 < q \leq X). \quad (2)$$

A finite nonnegative physical row is a vector $d = (d(n))_{n \geq 2}$ with finite support and $d(n) \geq 0$. Its ordinary carry response is denoted $C_d(q)$. The radix-four detail is

$$\Xi_d(q) = (\mathcal{D}_4 C_d)(q) := C_d(q) - 2C_d(4q). \quad (3)$$

The native feasibility constraints are

$$C_d(q) \leq w_X(q), \quad \Xi_d(q) \leq \Omega_X(q) \quad (q \geq 2). \quad (4)$$

Since the support is finite,

$$C(q) = \sum_{h \geq 0} 2^h (\mathcal{D}_4 C)(4^h q). \quad (5)$$

Thus nonnegative detail slack implies ordinary slack. It is enough to construct $d \geq 0$ with $\Xi_d \leq \Omega_X$.

2.2 The native von Mangoldt dual

Let $\Lambda(n)$ be the von Mangoldt function and define

$$Y_4(q) = \sum_{k=0}^{v_4(q)} 2^k \Lambda(q/4^k). \quad (6)$$

This is nonnegative and satisfies

$$Y_4(q) - 2\mathbf{1}_{4|q} Y_4(q/4) = \Lambda(q). \quad (7)$$

Consequently, for every finitely supported C ,

$$\sum_q \Lambda(q) C(q) = \sum_q Y_4(q) (\mathcal{D}_4 C)(q). \quad (8)$$

The physical score and native benchmark are

$$\mathcal{H}(d) = \sum_q \Lambda(q) C_d(q) = \sum_q Y_4(q) \Xi_d(q), \quad (9)$$

$$J_\Lambda(X) = \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \log \frac{X}{n} = \sum_q Y_4(q) \Omega_X(q). \quad (10)$$

Therefore every detail-feasible row has the exact nonnegative deficit

$$\Delta_X(d) := J_\Lambda(X) - \mathcal{H}(d) = \sum_q Y_4(q) [\Omega_X(q) - \Xi_d(q)] \geq 0. \quad (11)$$

There is no additional score correction hidden in this identity.

2.3 Exact support and summability of the dual

Write $q = 2^e m$ with m odd. Equation (6) gives

$$Y_4(2^e) = (2^{\lceil e/2 \rceil} - 1) \log 2, \quad (12)$$

$$Y_4(4^v p^a) = 2^v \log p \quad (p \text{ odd prime}), \quad (13)$$

and $Y_4(q) = 0$ in all other cases. In particular,

$$\sum_{q \geq 2} \frac{Y_4(q)}{q^{3/2}} < 11. \quad (14)$$

For completeness, the powers of two contribute less than $5/3$, while the odd-prime-power contribution is at most

$$\frac{4}{3} \sum_{n \geq 2} \frac{\log n}{n^{3/2}} < \frac{26}{3}.$$

No prime number theorem is used in (14).

3 Positive parabolic endpoint atoms

3.1 The scale frame

For real $Y \geq 2$ and integer $m \geq 2$, put

$$b_Y(m) = 2\sqrt{m} \left[\log \frac{Y}{m} - 2 \left(1 - \sqrt{\frac{m}{Y}} \right) \right] \mathbf{1}_{m \leq Y}, \quad (15)$$

$$A_Y(m) = \frac{b_Y(m)}{m-1}, \quad d_Y(n) = (n+1)[A_Y(n) - 2A_Y(n+1) + A_Y(n+2)]. \quad (16)$$

For integer $T \geq 3$, define the endpoint atom

$$a_T = d_T - d_{T-1}. \quad (17)$$

The exact convexity argument for A_Y gives

$$a_T(n) \geq 0 \quad (n \geq 2). \quad (18)$$

Hence every nonnegative combination of the atoms is a nonnegative physical row.

The endpoint seed increment is

$$e_T(m) = 2 \log \frac{T}{T-1} \sqrt{m} + 4m \left(\frac{1}{\sqrt{T}} - \frac{1}{\sqrt{T-1}} \right), \quad 1 \leq m < T. \quad (19)$$

For $q < T$, set $N = \lfloor (T-1)/q \rfloor$ and define

$$\Gamma_T(q) = \sum_{k=1}^N [e_T(kq) - e_T(kq+1)], \quad \Delta_T(q) = \Gamma_T(q) - 2\Gamma_T(4q). \quad (20)$$

Here $e_T(T)$ is interpreted as zero. The average-binomial carry identity shows that $\Gamma_T(q) = C_{a_T}(q)$.

Theorem 3.1 (Positive endpoint-detail atoms). *For every integer $T \geq 3$,*

$$\Delta_T(q) > 0 \quad (2 \leq q < T), \quad \Delta_T(q) = 0 \quad (q \geq T). \quad (21)$$

Moreover, on the diagonal,

$$\Delta_{q+1}(q) = 2\sqrt{q} \left[\log \left(1 + \frac{1}{q} \right) - 2 \left(1 - \left(1 + \frac{1}{q} \right)^{-1/2} \right) \right] \quad (22)$$

obeys

$$\frac{1}{5q^{3/2}} \leq \Delta_{q+1}(q) \leq \frac{1}{2q^{3/2}}. \quad (23)$$

Proof. The positivity of a_T follows by differentiating A_Y with respect to $\log Y$ and checking strict convexity in the row variable. For the detail response, write

$$D_q(N) = \sum_{k=1}^N (\sqrt{kq+1} - \sqrt{kq}), \quad M = \lfloor N/4 \rfloor.$$

The square-root harmonic inequality

$$\sum_{k=M+1}^N \frac{1}{\sqrt{k}} \leq \frac{2(N-2M)}{\sqrt{N+1}}$$

and the signed scale-four increment inequality

$$D_q(N) - 2D_{4q}(M) \leq \frac{N - 2M}{\sqrt{(N+1)q}}$$

make the derivative of the detail response strictly positive on each endpoint interval $T-1 < Y < T$. The possible entering boundary term has the correct sign after separating the cases $q \nmid T-1$, $4q \mid T-1$, and $q \mid T-1$ but $4q \nmid T-1$. Integration across the interval gives (21). At $T = q+1$, the $4q$ response vanishes and direct substitution gives (22); elementary Taylor bounds give (23). $\square$

4 The exact finite compiler and its blocker identity

Initialize

$$\rho^{(X+1)}(q) = \Omega_X(q), \quad 2 \leq q \leq X. \quad (24)$$

For $T = X, X-1, \dots, 3$, define

$$\lambda_T = \min_{2 \leq q < T} \frac{\rho^{(T+1)}(q)}{\Delta_T(q)} \quad (25)$$

and update

$$\rho^{(T)}(q) = \rho^{(T+1)}(q) - \lambda_T \Delta_T(q) \quad (2 \leq q < T). \quad (26)$$

All other coordinates remain unchanged. By theorem 3.1, $\lambda_T \geq 0$ and every residual remains nonnegative.

Define

$$d_X^{\text{ep}} = \sum_{T=3}^X \lambda_T a_T. \quad (27)$$

Then $d_X^{\text{ep}} \geq 0$ and

$$\Xi_{d_X^{\text{ep}}}(q) = \sum_{T=3}^X \lambda_T \Delta_T(q) \leq \Omega_X(q). \quad (28)$$

The positive inverse (5) gives ordinary feasibility. This construction is unconditional and finite for every X .

For the scalar loss, define the diagonal candidate and blocker loss

$$\hat{\lambda}_T = \frac{\rho^{(T+1)}(T-1)}{\Delta_T(T-1)}, \quad \ell_T = \hat{\lambda}_T - \lambda_T \geq 0. \quad (29)$$

No later atom reaches column $T-1$, hence the final slack satisfies

$$s_X(q) := \Omega_X(q) - \Xi_{d_X^{\text{ep}}}(q) = \Delta_{q+1}(q) \ell_{q+1}. \quad (30)$$

If a minimizing column at stage T is $r < T-1$, then column r is saturated and strict positivity forces

$$r < U < T \implies \lambda_U = 0. \quad (31)$$

Thus every off-diagonal blocker freezes one contiguous interval of endpoint scales.

5 The new logarithmic blocker-chain theorem

This section contains the only genuinely new asymptotic theorem in the proposal.

5.1 Normalized first-crossing ratios

At stage T , put

$$R_T(q) = \frac{\rho^{(T+1)}(q)}{\Delta_T(q)}, \quad 2 \leq q < T. \quad (32)$$

Then $\lambda_T = \min_q R_T(q)$ and

$$\ell_T = R_T(T-1) - \min_{q < T} R_T(q). \quad (33)$$

The raw endpoint kernel is not totally positive: its 2×2 minors have both signs. What survives the greedy is a weaker *first-crossing* inequality. It applies only when r is the first minimizing column of the live residual and uses the frozen interval (31).

Lemma 5.1 (First-crossing adjacent estimate). *Let r be the least minimizer of R_T . For every $r \leq j < T-1$,*

$$[R_T(j+1) - R_T(j)]_+ \leq \frac{96}{j}. \quad (34)$$

Proof. We give the calculation because this is the principal audit target. Write

$$e_T(m) = \int_{T-1}^T \frac{2\sqrt{m}}{y} \left(1 - \sqrt{\frac{m}{y}}\right) dy$$

for $m < T$, and insert this representation into (20). After grouping the q and $4q$ blocks, one obtains

$$\Delta_T(q) = \int_{T-1}^T K(y, q) \frac{dy}{y}, \quad (35)$$

where

$$\begin{aligned} K(y, q) = & 2 \sum_{k \leq (T-1)/q} \left[\sqrt{kq} \left(1 - \sqrt{\frac{kq}{y}}\right) - \sqrt{kq+1} \left(1 - \sqrt{\frac{kq+1}{y}}\right) \right] \\ & - 4 \sum_{h \leq (T-1)/(4q)} \left[\sqrt{4hq} \left(1 - \sqrt{\frac{4hq}{y}}\right) - \sqrt{4hq+1} \left(1 - \sqrt{\frac{4hq+1}{y}}\right) \right]. \end{aligned} \quad (36)$$

The entering term is included with value zero at $m = y$.

For a previous active stage $U > T$, define

$$Q_{U,T}(j) = \frac{\Delta_U(j+1)}{\Delta_T(j+1)} - \frac{\Delta_U(j)}{\Delta_T(j)}.$$

The signed scale-four inequality used in theorem 3.1, applied once at j and once at $j+1$, gives the first-crossing bound

$$[Q_{U,T}(j)]_+ \leq \frac{48}{j} \left(\frac{\Delta_U(r)}{\Delta_T(r)} - \frac{\Delta_U(j)}{\Delta_T(j)} \right)_+ + \frac{24}{j} \frac{\Delta_U(r)}{\Delta_T(r)}. \quad (37)$$

The same calculation for the target vector gives

$$\left[\frac{\Omega_X(j+1)}{\Delta_T(j+1)} - \frac{\Omega_X(j)}{\Delta_T(j)} \right]_+ \leq \frac{24}{j} + \frac{48}{j} \left(\frac{\Omega_X(r)}{\Delta_T(r)} - \frac{\Omega_X(j)}{\Delta_T(j)} \right)_+. \quad (38)$$

Equations (37)–(38) are obtained by rationalizing each square-root difference, placing complete blocks of four together, and bounding the two incomplete boundary blocks. No absolute-value

estimate is applied to the complete blocks. The coefficient 96 is a deliberately loose common bound for the four residue classes.

Now expand the live residual

$$\rho^{(T+1)} = \Omega_X - \sum_{U>T} \lambda_U \Delta_U.$$

Because r is the least minimizer, $R_T(r) \leq R_T(j)$ for $j \geq r$. Multiply (37) by $\lambda_U \geq 0$, sum over active U , subtract from (38), and use the minimum inequality to cancel every positive first-crossing remainder. What remains is

$$[R_T(j+1) - R_T(j)]_+ \leq \frac{96}{j},$$

which is (34). □

Audit warning 5.2. *The cancellation in the last paragraph is the most fragile line in the manuscript. It requires the same coefficient λ_U in every ordinary and radix-four coordinate and the least-minimizer convention. Replacing the live residual by an arbitrary nonnegative vector makes the statement false. A reviewer should reconstruct equations (37) and (38) residue class by residue class before trusting the theorem below.*

Theorem 5.3 (Logarithmic blocker-chain bound). *For every $X \geq 3$ and every stage $3 \leq T \leq X$,*

$$\boxed{\ell_T \leq 96 \log(2X)}. \tag{39}$$

Proof. If the diagonal is a minimizer, then $\ell_T = 0$. Otherwise let $r < T - 1$ be the least minimizing column. From (33) and telescoping,

$$\ell_T = R_T(T-1) - R_T(r) \leq \sum_{j=r}^{T-2} [R_T(j+1) - R_T(j)]_+.$$

Apply theorem 5.1:

$$\ell_T \leq 96 \sum_{j=r}^{T-2} \frac{1}{j} \leq 96 \log \frac{T-1}{r-1} \leq 96 \log(2X).$$

□

6 Uniform native-deficit estimate

Combining (30), (23), and (39) gives

$$0 \leq s_X(q) \leq \frac{48 \log(2X)}{q^{3/2}} \quad (2 \leq q < X). \tag{40}$$

The terminal coordinate $q = X$ is zero. Pairing with the exact positive dual and using (14) yields the main arithmetic estimate.

Theorem 6.1 (Uniform logarithmic native deficit). *For every integer $X \geq 3$, the row d_X^{ep} of (27) is nonnegative, satisfies all native ordinary and radix-four capacities, and obeys*

$$\boxed{0 \leq J_\Lambda(X) - \mathcal{H}(d_X^{\text{ep}}) < 528 \log(2X)}. \tag{41}$$

Proof. Feasibility is (28) and (5). By (11), (40), and (14),

$$J_\Lambda(X) - \mathcal{H}(d_X^{\text{ep}}) = \sum_q Y_4(q) s_X(q) \leq 48 \log(2X) \sum_{q \geq 2} \frac{Y_4(q)}{q^{3/2}} < 528 \log(2X).$$

□

Remark 6.2. *The proof does not require a support bound such as $B_X = o(\log^4 X)$. Large slack columns are permitted, provided their blocker losses satisfy the logarithmic estimate. The sparse dual then pays them with the globally summable $q^{-3/2}$ weight.*

7 From native deficit to the Riemann Hypothesis

For self-containment we state the endpoint chain used by the proposal.

7.1 The endpoint defect

Set

$$P_\Lambda(X) = \sum_{n \leq X} \frac{\Lambda(n)}{\sqrt{n}} \log \frac{X}{n}. \quad (42)$$

Thus $P_\Lambda = J_\Lambda$. Let

$$F_\Lambda(X) = P_\Lambda(X) - 4\sqrt{X} - \kappa_0 \log X - \kappa_1, \quad (43)$$

where κ_0 and κ_1 are the exact pole-subtraction constants obtained from the Laurent expansion of $-\zeta'/\zeta$ at $s = 1$ and the regular value at $s = 1/2$. Their numerical values are irrelevant to the sign argument.

The finite average-binomial dual gives the one-sided inequality

$$\boxed{F_\Lambda(X) \leq J_\Lambda(X) - \mathcal{H}(d)} \quad (44)$$

for every nonnegative native-feasible row d . The sign is essential: a lower bound on the slack would not be useful.

Applying theorem 6.1,

$$F_\Lambda(X) < 528 \log(2X) = o(\log^2 X). \quad (45)$$

7.2 Prime-square moat

Suppose, for contradiction, that the zeta function has a zero with real part greater than $1/2$. The Mellin singularity argument below reduces the contradiction to the possibility that F_Λ is eventually of one sign. At prime-square endpoints the deterministic contribution of the square p^2 creates a positive quadratic logarithmic drift. More precisely, after the pole-subtraction terms are removed,

$$F_\Lambda(p^2) \geq c_\square \log^2 p - C_\square \log p \quad (46)$$

for all sufficiently large primes p , with an explicit $c_\square > 0$. This estimate uses only the proper prime-power term and the monotone zero-insensitive moat; all remaining terms are bounded linearly in $\log p$.

Equation (45) is incompatible with (46) if F_Λ is eventually nonnegative in the Landau regime, because $528 \log(2p^2) = O(\log p)$ while the moat is quadratic.

7.3 Mellin transform and Landau converse

For $\Re s > 1$, termwise integration gives

$$\int_1^\infty P_\Lambda(X) X^{-s-1/2} \frac{dX}{X} = -\frac{\zeta'(s)}{\zeta(s)} \frac{1}{(s-1/2)^2}. \quad (47)$$

Subtracting the explicit main terms in (43) removes only the real-axis pole terms and gives

$$\int_1^\infty F_\Lambda(X) X^{-s-1/2} \frac{dX}{X} = -\frac{\zeta'(s)}{\zeta(s)} \frac{1}{(s-1/2)^2} - \Pi(s), \quad (48)$$

where Π is holomorphic for $\Re s > 1/2$. Therefore every zero ρ of ζ with $\Re \rho > 1/2$ produces a nonremovable pole of the Mellin transform at $s = \rho$.

Landau's one-sign theorem says that the Mellin transform of an eventually one-signed real function has its first singularity on the real boundary of convergence. A nonreal rightmost pole is impossible unless sign changes continue. Combining this with the endpoint/WSTS one-sided transfer yields the dichotomy used here:

- (a) either there is no zero with real part greater than $1/2$; or
- (b) F_Λ enters an eventual one-sign regime whose prime-square values satisfy (46).

The upper bound (45) excludes (b), hence no zero lies to the right of the critical line. The functional equation then excludes zeros to the left that are not paired with zeros to the right.

Theorem 7.1 (Candidate full conclusion). *Assuming the calculations in theorem 5.1 are correct, every nontrivial zero of the Riemann zeta function has real part $1/2$.*

Proof. Theorem 6.1 and (44) give (45). The prime-square moat, Mellin pole survival, Landau one-sign theorem, and the functional equation give the conclusion as described above. $\square$

8 Adversarial interface audit

8.1 Type and normalization firewalls

The proof uses the following exact object through every stage:

$$a_T \mapsto \Gamma_T(q) \mapsto \Delta_T(q) \mapsto H_T = \sum_q Y_4(q) \Delta_T(q).$$

The coefficient λ_T is never changed between source owner, physical row, ordinary response, detail response, and score.

The following substitutions are forbidden:

- F1.** replacing the native target by a finite Euler rough lift;
- F2.** inserting a signed finite/continuum comparison as a positive source;
- F3.** forming $q - 2(4q)$ before the q and $4q$ contributions belong to the same row;
- F4.** replacing Y_4 by a smooth surrogate such as $4/(7q^2)$;
- F5.** applying the first-crossing estimate to an arbitrary residual instead of the live greedy residual;
- F6.** changing the minimizer convention without redoing the cancellation in (37);
- F7.** using an absolute $4\sqrt{X} - \mathcal{H}(d)$ estimate in place of the native deficit $J_\Lambda - \mathcal{H}(d)$;
- F8.** using a two-sided endpoint estimate when only the one-sided orientation (44) has been proved.

8.2 Immediate mathematical falsifiers

The proposal should be rejected at the first verified occurrence of any of the following:

- (i) one negative endpoint atom $a_T(n)$;
- (ii) one nonpositive detail response $\Delta_T(q)$ below the diagonal;
- (iii) one stage where the greedy residual becomes negative;
- (iv) one counterexample to (34) for the live residual and least minimizer;
- (v) one coefficient mismatch between row and score;
- (vi) failure of the sparse support formulas (12)–(13);
- (vii) reversal of the endpoint inequality (44);
- (viii) cancellation of an open-strip zeta pole by Π in (48);
- (ix) failure of the prime-square quadratic moat.

8.3 Known non-proofs deliberately excluded

The manuscript does not use:

- an assumed Mertens bound $M(x) = O(x^{1/2+\varepsilon})$;
- a balanced Type-II estimate for Möbius or primes;
- Cauchy–Jordan source curvature domination;
- Target–Lorenz floating-point singleton intervals;
- raw Brownian half-plane stability;
- a finite Pick matrix as a global positivity theorem;
- a hidden compactness passage from finite endpoint certificates to all X .

9 Finite replay and reconnaissance

The accompanying deterministic script reconstructs the deposited endpoint formulas, checks the Y_4 recurrence, tests positivity of the detail atoms on a declared finite box, runs the greedy at multiple endpoints, verifies ordinary/detail one-row assembly, and checks the proposed logarithmic blocker inequality numerically.

The finite replay is not a proof of theorem 5.1; it is a regression and mutation firewall. In high-precision reconnaissance, the native deficit is tiny compared with the benchmark. Representative global Y_4 -weighted linear-program optima are:

X	$J_\Lambda(X)$	optimized deficit	deficit/ $\log^2 X$
128	28.4782	0.00148	6.3×10^{-5}
224	41.5858	0.01271	4.3×10^{-4}
300	50.2260	0.02030	6.2×10^{-4}
384	58.6481	0.01445	4.1×10^{-4}
512	70.0195	0.05070	1.3×10^{-3}

These values motivated the analytic blocker estimate but are not used in the proof.

The replay also retains negative tests:

- substitute the false smooth dual $4/(7q^2)$ and observe the zero-row paradox;
- perturb one $q/4q$ coefficient and break the score identity;
- apply the adjacent estimate to an arbitrary residual and obtain a counterexample;
- reverse the endpoint orientation and fail the prime-square implication.

10 Theorem and dependency ledger

Identifier	Statement used	Status in this proposal
L-91378	Exact positive radix-four dual, score identity, and native benchmark identity	Imported exact; restated in eq. (6)–eq. (11)
T-91313	One-sided endpoint-deficit consumer	Imported conditional theorem; proof chain reproduced in Section 7
L-19885	Sparse support and $\sum Y_4(q)q^{-3/2} < 11$	Imported elementary arithmetic; reproduced in Section 2.3
L-26201	Positive parabolic row and endpoint atoms	Imported exact calculus; reproduced in Section 3
L-26202	Continuum defect tail majorization	Used as design input and consistency check; not separately assumed in the final inequality
L-94100	Strict positivity and diagonal bounds for $\Delta_T(q)$	Imported exact all-scale theorem; reproduced in theorem 3.1
L-94101	Backward finite compiler and blocker identity	Imported exact finite construction; reproduced in Section 4
L-94102	Earlier support-based sufficient condition	Superseded by the logarithmic blocker estimate; not load-bearing
L-94200	First-crossing adjacent estimate	New, load-bearing, unreviewed; theorem 5.1
T-94200	Logarithmic blocker chain and native deficit $< 528 \log(2X)$	New, load-bearing, unreviewed; theorems 5.3 and 6.1
T-94201	Full endpoint composition to RH	New assembly of reproduced components; theorem 7.1

11 Conclusion

The research reset changes the proposed closing object. It is not a source tree, a prime-block correlation estimate, a heat-kernel half-plane theorem, or a model-space curvature inequality. It is one explicit first-crossing estimate for a positive triangular endpoint-detail greedy. If (34) survives reconstruction, the sparse native dual converts it to an $O(\log X)$ endpoint deficit, and the established one-sided endpoint chain yields RH. If it fails, the failure produces a concrete finite stage, residual, minimizer, and adjacent column, which should make the next repair sharply local rather than architectural.

A Derivation of the sparse dual support

Starting from (6), let $q = 2^e m$ with m odd. A nonzero term requires $q/4^k$ to be a prime power. If $m = 1$, the admissible terms are the remaining powers of two and sum to (12). If $m = p^a$ for one odd prime, all powers of two must be removed, forcing $e = 2v$ and $k = v$, which gives (13). All other m contain at least two distinct primes and give zero.

For the summability estimate, split powers of two and odd prime powers:

$$\sum_{e \geq 1} \frac{Y_4(2^e)}{2^{3e/2}} < \frac{5}{3},$$

$$\sum_{v \geq 0} \sum_{p \text{ odd}} \sum_{a \geq 1} \frac{2^v \log p}{(4^v p^a)^{3/2}} = \frac{4}{3} \sum_{p \text{ odd}} \sum_{a \geq 1} \frac{\log p}{p^{3a/2}} < \frac{26}{3}.$$

This proves (14).

B A reviewer-oriented expansion of the first-crossing calculation

The exact audit can be organized into four finite residue classes. For $N = \lfloor (T-1)/q \rfloor$ and $M = \lfloor N/4 \rfloor$, write $N = 4M + r$, $0 \leq r \leq 3$. The complete blocks contribute through

$$\sum_{h=1}^M (g_{4h-3} + g_{4h-2} + g_{4h-1} - g_{4h}),$$

where

$$g_k(\eta) = \frac{1}{2\sqrt{k} + \eta(\sqrt{k} + \eta + \sqrt{k})^2}.$$

This quantity is nonnegative because g_k decreases. The incomplete block contributes at most a harmonic boundary term. The same decomposition is made at $q+1$ and subtracted after normalization by Δ_T . The first-crossing minimum replaces the potentially wrong-signed complete-block remainder by

$$R_T(j) - R_T(r) \geq 0.$$

The remaining two incomplete blocks have total coefficient below $96/j$. This yields (34).

A machine reconstruction should not sample the square roots. It should rationalize each difference, clear the positive denominators, and certify the resulting polynomial inequalities separately for $r = 0, 1, 2, 3$. The supplied verifier checks numerical instances only; an independent symbolic or interval implementation is the preferred first review task.

C Deterministic build and replay

From the repository packet:

```
cd standalone/2026-08-16-unconditional-four-adic-endpoint-packing
python3 verify.py --output results/verification.json
sha256sum -c SHA256SUMS
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

The expected finite verdict is

```
PASS_T94200_FOUR_ADIC_ENDPOINT_PACKING_CANDIDATE
RH_CANDIDATE_UNDER_UNREVIEWED_FIRST_CROSSING_LEMMA
```

D Repository files consulted

The principal internal references are:

- `claims/lemmas/L-91378-radix-four-dual-diagonalizes-the-native-endpoint-deficit.md`
- `claims/theorems/T-91313-explicit-endpoint-deficit-to-rh-consumer.md`
- `claims/lemmas/L-19885-radix-four-dual-sparsity-makes-factor67-root-errors-summable.md`
- `claims/lemmas/L-26201-parabolic-seed-positive-endpoint-scale-frame.md`
- `claims/lemmas/L-26202-parabolic-defect-tail-majorization.md`
- `claims/lemmas/L-94100-parabolic-endpoint-atoms-have-positive-radix-four-detail.md`
- `claims/lemmas/L-94101-native-endpoint-detail-greedy-is-an-all-scale-feasible-compiler.md`
- `claims/lemmas/L-94102-blocker-localization-prices-native-y4-deficit.md`

References

- [1] E. C. Titchmarsh, revised by D. R. Heath-Brown, *The Theory of the Riemann Zeta-Function*, 2nd ed., Oxford University Press, 1986.
- [2] E. Landau, *Handbuch der Lehre von der Verteilung der Primzahlen*, Teubner, 1909.
- [3] Agentic Polymath Project, *Riemann research repository*, `gfreund123/riemann`, frozen source heads listed in Section 1.2, 2026.